

Vacuum cleaner with bellows, 1910



Vacuum cleaners

By the middle of the 19th century, scientists understood how vacuum cleaning might work—by creating a partial vacuum within a device so as to suck up dirt and dust. However, it took until the early 20th century for an inventor to design a contraption that worked well enough to become a hit with the public.

EARLY CLEANERS

The American inventor Daniel Hess came up with the first dust-sucking machine for the home in 1860. It consisted of a wheeled carpet sweeper with a rotating brush that was positioned below a device called a bellows (a squeezable air bag), which could be pushed up and down to create a vacuum. This invention was not a success.

POWERED VACUUM CLEANERS

The game changer arrived in 1901 when the British engineer Hubert Cecil Booth created a dust-eating monster, nicknamed “Puffing Billy.” Powered by a gasoline engine, it weighed 4,000 lb (1,800 kg) and had to be transported by horse-drawn carts. It did not have brushes, but worked by sucking up dust through long tubes.

▼ PUFFING BILLY

Booth's machine was too big to be brought inside a house. Instead, it was parked on the street while hoses were pushed through the windows to suck up dust.

